Fiscal Year:	FY 2013	**Task Last Updated:**	FY 11/08/2012
PI Name:	Zhu, Xiongwei Ph.D.		
Project Title:	Two-Hit Model of Alzheimer's Disease		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106-3860	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
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No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	no change		
COI Name (Institution):	Casadesus Smith, Gemma (Case Western Reserve University) Lee, Hyoung-gon (Case Western Reserve University) Wang, Xinglong (Case Western Reserve University)		
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Task Description:	Future missions in space will involve long duration travel beyond the magnetic field of the Earth. During these long-duration spaceflight, astronauts will be exposed to types and doses of radiation (cosmic rays) that are not experienced in low earth orbit. The hazards associated with the space radiation will likely impact various organs and systems and there is a persistent effect of exposure to HZE particles lasting beyond the time of exposure. Space radiation risk to the central nervous system (CNS) is a major concern to NASA. It is known that space-like radiation induces neuronal and behavioral deficits, including hippocampus-dependent cognitive impairments which are also features in Alzheimer's Disease (AD) patients. Indeed, there are similarities between the effects of radiation and AD including cognitive deficits and neurochemical abnormalities which calls for a detailed investigation on the potential risk for AD of space radiation. In this proposal, we aim at exploring the potential mechanisms by which exposure to HZE particles and proton leads to the development of long-term degenerative processes in neuronal function and cognitive performance related to AD and the relationship of these effects to radiation parameters (particle, dose, and LET) and individual characteristics (age and gender) in AD model based on a novel Two Hit Hypothesis of AD pathogenesis. Two specific aims are proposed: Aim 1: To determine whether space radiation induces AD-relevant pathological changes in fully differentiated primary hippocampal neurons from CRN8 APP Tg mouse and 3xTg AD mouse models. Aim 2: To determine whether space radiation induces neurodegeneration and exacerbate AD-relevant pathological/cognitive/behavioral changes in 3xTg AD mice. These proposed studies will specifically address many of the critical issues NASA has identified.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Detrimental effects of space irradiation on the CNS have been known for some time and emerging evidence supports a potential risk for AD as a valid concern for long duration space exploration such as lunar mission or a Mars mission. Based on our study on mitotic and oxidative signaling and the recent development in AD field, we proposed a novel Two Hit Hypothesis stating that both oxidative stress and aberrant mitotic stimuli can independently initiate, yet both are necessary to propagate disease pathogenesis and progression. Our hypothesis predicts that after certain threshold of exposure, space irradiation will cause increased oxidative stress in genetically predisposed individuals who are under mitotic stress and induce full blown AD. Therefore, our two hit hypothesis not only calls for a detailed examination of AD risk during space radiation exposure, if confirmed, but also provide the potential mechanisms and working model to estimate risk for space crew to develop AD.
Task Progress:	This is the second year of our project. We finished irradiation for the pilot study in the summer and fall run of the first year and finished collection of brain samples and analysis of pathology in the second year. We have participated in the summer and fall run in 2012 and irradiated 20 groups (12/group) of animals in the total of 240 mice. We also performed in vitro study to evaluate the effectiveness of various antioxidants and cell cycle inhibitors in rescuing neuronal death after space irradiation. For aim 1, to better define the temporal sequence of events, primary hippocampal neurons (DIV14) isolated from CRND8 mice (APP neurons) and WT mice (WT neurons) were exposed to 56Fe (600Mev/u) or 28Si (300Mev/u) at 0.5 Gy and observed in real time up to 48 hrs post-irradiation to study the production of reactive oxygen species by monitoring DCF fluorescence signal in living cells. Unlike the oxidative damage marker 8-OHG we used before which demonstrated significant elevation in irradiated cells 24hr after irradiation, real time monitoring suggested that ROS signal in fact increase as early as 6 hrs, suggesting that oxidative stress is also an early event in this model. This study also confirmed increased basal oxidative stress and increased oxidative stress after irradiation in the APP transgenic neurons comparing to WT neurons. To determine the causal relationship between oxidative stress and cell cycle alterations and the subsequent cell death, we pretreated primary neuronal cells with various cell cycle inhibitors and/or antioxidants. Primary hippocampal neurons were incubated with these inhibitors during and after irradiation and harvested at various time points post-irradiation. Effects of these inhibitors were confirmed by immunofluoresence study of oxidative stress and cell cycle markers. Interestingly, cell cycle inhibitors such as flavopiridol demonstrate potent inhibition effect on irradiation-induced DNA damage (i.e., gammaH2AX signal) in both control and APP transgenic neurons. Antioxidants such as NAC also inhibits gammaH2AX signal in APP transgenic neurons, although to a lesser extent. However, NAC had no protective effects in irradiated WT neurons. Importantly, both cell cycle inhibitors and antioxidants demonstrated protection in cell death induced by irradiation in the APP transgenic neurons. In the fall run, we tried to replicate these findings and planned to explore the potential DNA repair pathways and stress activated kinase pathways involved by using specific inhibitors. However, primary neurons were contaminated during this round. We will try again in 2013. For aim 2, we collected and analyzed all the brain tissues for amyloid pathology from irradiated mice of the pilot study. In the pilot study, APP transgenic mice were exposed to 56Fe (600Mev/u) and 28Si (300Mev/u) at 0.5 and 2 Gy at 3 and 6 months of age and sacrificed at various post-irradiation time. Mouse brains were harvested after cervical dislocation when the brains were rapidly removed. The brain was hemisected across the sagittal midline. One half of the brain will be drop fixed and processed to paraffin. Six-micrometer-thick sections were prepared in silane-coated slides and analyzed for amyloid pathology by immunocytochemical staining with two different Abeta antibodies. Total area occupied by amyloid plaques were measured and divided by the total hippocampal or cortical brain areas to calculate the plaque density in these brain areas as a quantitative index for the amyloid pathology. Our detailed analysis revealed that both 56Fe and 28Si at both dosages caused a trend towards increased hippocampal/cortical amyloid plaque density and accelerated accumulation of amyloid plaques in CRND8 transgenic mice irradiated at either 3 month or 6 month old. Nevertheless, it must be cautioned that most of the difference between the irradiated and non-irradiated groups did not reach significance because of the small n number used per group and the loss of mice during the course of the experiments, so these observations need to be substantiated by larger cohorts. We also performed Golgi staining and the quantitative analysis of golgi stain-impregnated segments of apical dendritic branches of the CA1 pyramidal neurons revealed that exposure to 56Fe and 28Si caused significant reduction in spine numbers which suggest a likely impairment in the synaptic function. Overall, this pathology-based pilot study suggest that 56Fe and 28Si at both 0.5Gy and 2Gy cause increased amyloid pathology in both hippocampus and cortex. This pilot study support the more detailed full cognitive/behavioral study. For the full cognitive/behavioral study, we planned to follow the same scheme of the pilot study with an increased n number of 12/group. Mice will be longitudinally subject to novel object recognition test once every month for the first three months post-irradiation and will also be subject to fear conditioning once per three months. We had planned to irradiate these mice in the summer and fall run in 2012. The summer run was finished successfully. However, one third

	of the mice (22 out of 66 mice) did not survive the shipment from BNL to CWRU after irradiation with 28Si during the fall run. Since we have successfully finished three runs before this accident, and had only 9 mice died on arrival out of a total of 142 mice, this unusual animal mortality is unlikely due to the space irradiation they received in BNL, and we believe something occurred during shipment (like excessive temperature) that put increased stress on the mice causing this loss. We will need to include more mice to account for this loss and the whole irradiation scheme will be finished by summer run in 2013. To monitor the shipping condition, we will also include proper equipment during shipment to record the conditions.
Bibliography Type:	Description: (Last Updated: 04/11/2018)
Abstracts for Journals and Proceedings	Wang X, Richardson S, Casadesus G, Lee HG, Zhu X. "Space-Like Radiation Causes Increased Alzheimer-like Pathology in CRND8 APP Transgenic Mice." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Wang X, Richardson S, Casadesus G, Lee HG, Zhu X. "Increased Neuronal Vulnerability to Space-Like Radiation in Primary Neuronal Model of Alzheimer's Disease." 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7097.pdf , Sep-2011